

TOPS TECH Task 3 (CS)

Task -3 Malware Analysis Basics Steps

1. **List and briefly describe each step involved in static malware analysis, using examples relevant to Android APKs or Windows EXE files you might encounter on your own device.**

Ans:

Static Malware Analysis – Steps Involved in Analyzing Malware Without Execution

Objective

- The objective of this task is to understand the process of Static Malware Analysis and learn the different steps used to examine a suspicious file without executing it.
- Static analysis helps cybersecurity professionals identify the characteristics, functionality, and potential risks of malware while ensuring that the analysis environment remains safe.
- This task also includes examples related to Android APK files and Windows EXE files commonly encountered during malware investigations.

Introduction

- Static Malware Analysis is the process of examining a suspicious file without executing or running it.
- Instead of observing the malware's runtime behavior, analysts inspect the file's structure, metadata, strings, digital signatures, imported libraries, and cryptographic hashes to understand its functionality.
- Static analysis is considered the first step in malware investigation because it is safe, fast, and does not require executing potentially harmful code.
- Security analysts use various tools to determine whether a file is malicious before performing dynamic analysis in a sandbox environment.

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Steps Involved in Static Malware Analysis

Step 1: File Identification

The first step is to identify the file type and its basic properties such as file name, extension, size, and format.

Purpose:

Determine whether the file is an EXE, DLL, APK, PDF, ZIP, or another format.

Verify if the extension matches the actual file type.

Example

Windows Example:

invoice.exe

Android Example:

WhatsApp_Update.apk

Step 2: Calculate File Hash

A cryptographic hash uniquely identifies a file. Common hash algorithms include MD5, SHA-1, and SHA-256.

Purpose:

Verify file integrity.

Compare the hash against malware databases such as VirusTotal.

Identify known malware samples.

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Example Hash

SHA-256:

3A7D9F56A91E5F3C2D847D5B4E63D7C8B46F4D6A17E2B5F1A7B8
9D8C4F9E1234

Step 3: Determine File Type

Security analysts verify the actual file format instead of trusting the file extension.

Purpose:

Detect disguised or renamed malware.
Confirm whether the file is executable.

Example

File Name:

photo.jpg.exe

Actual Type:

Windows Portable Executable (PE)

Step 4: Analyze Strings

String analysis extracts readable text embedded inside the file.

Examples include:

URLs

IP Addresses

File Paths

Registry Keys

User-Agent Strings

Error Messages

Purpose:

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Identify suspicious domains.
Detect hardcoded passwords.
Discover attacker infrastructure.

Example

`http://malicious-example.com`

`cmd.exe`

`powershell.exe`

`C:\Users\Public`

Step 5: Examine Metadata

Metadata contains information about the file.

Examples:

Creation Time
Compilation Date
Company Name
Product Version
Author Information

Purpose

Identify suspicious timestamps.
Detect fake software publishers.
Determine whether the file has been modified.
Step 6: Analyze PE Header (Windows EXE Files)

Portable Executable (PE) headers provide structural information about Windows executable files.

The PE header contains:

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Entry Point
Image Base
Imported DLLs
Exported Functions
Section Information

Purpose

Detect packed executables.
Identify abnormal sections.
Understand executable structure.
Step 7: Analyze Import Table

Windows executables import system libraries to perform specific tasks.

Common DLLs include:

kernel32.dll

user32.dll

advapi32.dll

ws2_32.dll

Purpose

Identify networking functions.
Detect file manipulation.
Discover registry operations.
Recognize process creation capabilities.
Step 8: Check Digital Signature

Many legitimate applications contain a valid digital signature.

Purpose

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Verify software authenticity.
Detect tampered executables.
Identify unsigned suspicious files.
Step 9: Scan Using VirusTotal

Upload the file hash or sample to VirusTotal.

VirusTotal compares the file against multiple antivirus engines.

Purpose

Detect known malware.
View detection names.
Review community analysis.
Identify Indicators of Compromise (IoCs).
Step 10: Document Findings

Record all collected information.

The report should include:

File Name
File Type
File Size
Hash Values
Suspicious Strings
Imported Libraries
VirusTotal Detection
Analyst Remarks

Proper documentation assists future investigations and incident response activities.

Example of Static Analysis
Windows EXE Example

File Name

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invoice.exe

File Type

Windows Portable Executable (PE)

SHA-256

3A7D9F56A91E5F3C2D847D5B4E63D7C8B46F4D6A17E2B5F1A7B8
9D8C4F9E1234

Imported DLLs

kernel32.dll

user32.dll

ws2_32.dll

VirusTotal Result

Detected by multiple antivirus engines.
Android APK Example

File Name

WhatsApp_Update.apk

Package Name

com.example.whatsapp

Permissions

READ_CONTACTS

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ACCESS_FINE_LOCATION

SEND_SMS

READ_SMS

Observation

The APK requests sensitive permissions that should be reviewed before installation.

Advantages of Static Malware Analysis

- Safe because the malware is never executed.
- Quick initial assessment.
- Detects known malware using file hashes.
- Reveals suspicious strings and metadata.
- Helps determine whether further dynamic analysis is required.
- Preserves the integrity of the analysis environment.

Limitations of Static Malware Analysis

- Cannot observe runtime behavior.
- Packed or encrypted malware may hide its code.
- Some malware uses obfuscation techniques.
- Unable to detect actions performed only during execution.
- Advanced malware may evade static analysis.

Result

- Static Malware Analysis was successfully studied by examining the different stages involved in analyzing suspicious files without executing them.
- The process included file identification, hash calculation, metadata analysis, string extraction, PE header inspection, import table analysis, digital signature verification, and VirusTotal scanning. These techniques help analysts identify potential threats while maintaining a secure analysis environment.

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Conclusion

- Static Malware Analysis is a fundamental cybersecurity technique used to investigate suspicious files safely without execution.
- It provides valuable information about file structure, metadata, cryptographic hashes, embedded strings, imported libraries, and digital signatures.
- Although static analysis cannot reveal runtime behavior, it serves as the foundation for malware investigation and complements dynamic analysis performed in sandbox environments. Combining both analysis techniques enables security professionals to accurately identify, classify, and respond to malware threats.

2. **Download a harmless sample file (such as the EICAR test file) and use Any.Run to perform a dynamic analysis. Record and explain at least three behaviors or indicators the tool detects.**
Hint: EICAR is a safe file used for testing antivirus tools and will not harm your system.

Ans:

Static Malware Analysis – Steps Involved in Analyzing Malware Without Execution

Objective

- The objective of this practical is to perform Dynamic Malware Analysis using the Any.
- Run Interactive Sandbox. A harmless EICAR test file is executed inside a secure sandbox environment to observe its runtime behavior.
- The purpose is to understand how sandbox environments monitor suspicious activities such as process execution, antivirus detection, and file operations without affecting the host operating system.

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Introduction

- Dynamic Malware Analysis is the process of executing a suspicious file inside a secure, isolated environment called a sandbox.
- Unlike Static Analysis, where the file is examined without execution, Dynamic Analysis observes the file's actual behavior during runtime.

Security analysts use dynamic analysis to monitor:

- Process creation
- File system activity
- Network communication
- Registry modifications
- Antivirus detection
- System behavior

For this practical, the EICAR Anti-Virus Test File is used. The EICAR file is not real malware but is intentionally designed to trigger antivirus detection for testing security software.

Software Used

- Any.Run Interactive Sandbox
- EICAR Test File
- Google Chrome
- Windows 10 / Windows 11

Practical Procedure

Step 1: Download the EICAR Test File

Download the harmless EICAR test file from the official EICAR website.

Example filename:

eicar.com

or

eicar_com.zip

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Step 2: Open Any.Run

Open the Any.Run Interactive Sandbox in your web browser.
Create a free account and log in.

Step 3: Upload the Sample

- Click New Task.
- Upload the downloaded EICAR file.
- Select the operating system (Windows 10).
- Start the analysis.

The sandbox safely executes the sample inside an isolated virtual machine.

Step 4: Observe the Runtime Behavior

During execution, Any.Run monitors:

- Running Processes
- File Activity
- Network Activity
- Detection Events
- Process Tree
- Indicators of Compromise (IoCs)

Observed Behaviors

Behavior 1: Antivirus Detection

- The sandbox immediately identified the EICAR file as a test malware sample.

Observation

- The file was detected by the antivirus engine using the standard EICAR Test File signature.

Security Importance

- This confirms that the antivirus detection engine is functioning correctly.

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Behavior 2: Process Execution

- The sandbox recorded the execution of the uploaded sample.

Observation

- A process associated with the uploaded file was created and monitored throughout the analysis session.

Security Importance

- Monitoring process execution helps analysts identify malicious child processes and suspicious application behavior.

Behavior 3: File Activity Monitoring

- The sandbox monitored all file operations performed during execution.

Observation

- The uploaded file was accessed and scanned by the analysis environment.
- No harmful modifications to the operating system were observed because the EICAR file is only a test sample.

Security Importance

- File activity monitoring helps detect malware attempting to create, modify, encrypt, or delete files.

Behavior 4: Network Activity

- The sandbox monitored network communication.

Observation

- No suspicious outbound network connections or command-and-control (C2) communication were detected.

Security Importance

- Real malware often contacts remote servers to download additional payloads or transmit stolen information. Monitoring network activity helps identify these behaviors.

Behavior 5: Indicators of Compromise (IoCs)

- The sandbox generated Indicators of Compromise related to the uploaded sample.

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Examples include:

- File Name
- File Hash
- Detection Label
- Process Information

These indicators assist security analysts during threat hunting and incident response.

Summary of Observed Behaviors

Behavior	Observation	Security Significance
Antivirus Detection	EICAR detected as a test file	Confirms antivirus functionality
Process Execution	Sample executed inside sandbox	Enables runtime behavior analysis
File Activity	File accessed and monitored	Detects unauthorized file modifications
Network Activity	No external communication observed	Confirms no malicious network behavior
Indicators of Compromise	Hash, filename, process information collected	Supports incident response and threat intelligence

Advantages of Dynamic Malware

- Observes real runtime behavior.
- Detects hidden malicious activities.
- Identifies process creation.
- Detects network communication.
- Monitors registry modifications.
- Reveals Indicators of Compromise (IoCs).
- Safe execution inside an isolated environment.

Limitations of Dynamic Malware Analysis

- Requires a sandbox environment.
- Some advanced malware detects virtual machines and changes its behavior.

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- Analysis may take longer than static analysis. Analysis
- Certain malware activates only under specific conditions or after long execution times.

Result

- The EICAR test file was successfully analyzed using the Any.
- Run Interactive Sandbox.
- During execution, the sandbox detected the test file, monitored its execution, analyzed file operations, and observed network activity.
- The analysis confirmed that the EICAR sample is a harmless antivirus testing file and demonstrated how dynamic malware analysis safely monitors file behavior within an isolated environment.

Conclusion

- Dynamic Malware Analysis provides valuable insight into the runtime behavior of suspicious files by executing them inside a controlled sandbox environment.
- In this practical, the EICAR test file was safely analyzed using Any.Run. The sandbox successfully monitored process execution, antivirus detection, file operations, and network activity without causing any harm to the host system.
- Dynamic analysis is an essential technique in modern cybersecurity because it helps analysts understand malware behavior and detect threats that may not be visible through static analysis alone.

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3. **Use Cuckoo Sandbox (online demo or local setup) to analyze a suspicious file of your choice (not the one used in class). Take screenshots of the process and summarize what network connections or file changes the sandbox reports.**

Ans:

Malware Analysis Using Cuckoo Sandbox

Objective

- The objective of this practical is to analyze a suspicious executable file using Cuckoo Sandbox, an automated malware analysis system.
- The analysis focuses on observing the file's runtime behavior, including process execution, file system modifications, registry changes, and network communication.
- This helps security analysts understand how malware behaves in a controlled and isolated environment without affecting the host system.

Introduction

- Cuckoo Sandbox is an open-source automated malware analysis platform designed to safely execute suspicious files inside an isolated virtual machine.
- It records every action performed by the file, including process creation, registry modifications, file operations, network communication, and system calls.
- Unlike Static Analysis, which only examines the file without executing it, Dynamic Analysis performed through Cuckoo Sandbox provides a detailed view of the malware's behavior during execution.
- Security analysts use these observations to identify malicious activities, create Indicators of Compromise (IoCs), and improve incident response.
- For this practical, a suspicious Windows executable file was analyzed in a sandbox environment.

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Summary of Sandbox Findings

Category	Observation
Process Activity	Executable created one or more processes during execution.
File Changes	Temporary files were created and modified.
Registry Activity	Registry keys were accessed and updated.
Network Connections	DNS and HTTP requests were observed.
Indicators of Compromise	Hash values, process information, registry keys, and network indicators were generated.

Advantages of Using Cuckoo Sandbox

- Safe malware execution in an isolated environment.
- Automated malware analysis.
- Detailed behavioral reports.
- Detects process creation.
- Monitors registry modifications.
- Captures network communication.
- Generates Indicators of Compromise (IoCs).
- Supports incident response and malware research.

Limitations

- Some advanced malware can detect virtual machines.
- Long-running malware may require extended analysis time.
- Certain malware activates only under specific user interactions.
- Results depend on the sandbox configuration.

Result

- The suspicious executable file was successfully analyzed using Cuckoo Sandbox.
- The sandbox monitored process execution, file system changes, registry modifications, and network activity.

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- The generated report provided valuable Indicators of Compromise that can assist cybersecurity analysts during malware investigations and threat detection.

Conclusion

- Cuckoo Sandbox is an effective malware analysis platform that enables safe execution and detailed observation of suspicious files.
 - In this practical, the uploaded executable was analyzed inside an isolated virtual environment without affecting the host operating system.
 - The analysis demonstrated how security professionals can identify malicious behaviors, monitor file and registry changes, observe network communications, and generate Indicators of Compromise for further investigation.
 - Dynamic analysis using Cuckoo Sandbox plays a critical role in modern malware research and incident response.
4. **Compare the results from static and dynamic analysis for the same file you analyzed in Any.Run or Cuckoo Sandbox. Write 3 differences in the type of information each method provides.**

Ans:

Comparison Between Static and Dynamic Malware Analysis

Objective

- The objective of this task is to compare Static Malware Analysis and Dynamic Malware Analysis based on the information they provide while analyzing the same malware sample.

Introduction

- Malware analysis helps security professionals understand how malicious software works. There are two main approaches:
 - Static Analysis – The malware is analyzed without executing it.

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- **Dynamic Analysis** – The malware is executed inside a secure sandbox to observe its behavior.

Both techniques are important and complement each other during malware investigations.

Comparison Between Static and Dynamic Malware Analysis

Static Malware Analysis	Dynamic Malware Analysis
The file is analyzed without execution.	The file is executed inside a sandbox environment.
Provides information such as file type, hash values, metadata, and embedded strings.	Provides runtime behavior such as process creation, file changes, registry modifications, and network activity.
Safe, fast, and suitable for initial investigation.	More detailed but requires an isolated analysis environment.

Three Major Differences

1. File Execution

- **Static Analysis:** The malware is **not executed**.
- **Dynamic Analysis:** The malware is **executed safely** inside a sandbox.

2. Information Collected

- **Static Analysis:** Collects file properties, hashes, metadata, imports, and strings.
- **Dynamic Analysis:** Collects runtime behavior, process activity, registry changes, network traffic, and file modifications.

3. Purpose

- **Static Analysis:** Used for quick identification and initial investigation.
- **Dynamic Analysis:** Used to understand the actual behavior and impact of the malware.

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Result

- Both static and dynamic malware analysis provide valuable information for malware investigation.
- Static analysis offers quick insights into the file structure, while dynamic analysis reveals the malware's real-time behavior during execution.

Conclusion

- Static and Dynamic Malware Analysis complement each other in cybersecurity.
- Static analysis is useful for safely examining a file, whereas dynamic analysis helps analysts observe real-time malicious activities.
- Using both techniques together provides a complete understanding of malware and improves threat detection.

5. **Find a recent malware sample report on Any.Run's public submissions and identify at least two techniques the malware uses to avoid detection. Briefly explain how these techniques work.**

Ans:

Detection-Evasion Techniques Used by Malware (Based on Any.Run Report)

Objective

- The objective of this task is to study a recent malware analysis report from Any.
- Run and identify two detection-evasion techniques used by malware to avoid security tools and antivirus detection.

Introduction

- Modern malware often uses detection-evasion techniques to bypass antivirus software, sandbox environments, and security monitoring tools.

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- These techniques help attackers hide malicious activities and delay detection.
- Understanding these methods enables cybersecurity professionals to improve malware detection and incident response.

Detection-Evasion Techniques

1. Anti-Virtual Machine (Anti-VM) Detection

- Some malware checks whether it is running inside a virtual machine or sandbox environment. If a virtual machine is detected, the malware stops executing or hides its malicious behavior to avoid analysis.

Purpose:

- To prevent security researchers from analyzing the malware inside a sandbox.

2. Process Injection

- Process Injection is a technique in which malware injects its malicious code into a legitimate system process such as explorer.exe or svchost.exe. This allows the malware to hide its activity and bypass some security solutions.

Purpose:

- To execute malicious code under the identity of a trusted Windows process.

Importance of Detecting Evasion Techniques

- Helps security analysts identify hidden malware.
- Improves malware detection accuracy.
- Supports effective incident response.
- Strengthens endpoint security.

Result:

- The Any.Run malware report demonstrated that attackers use techniques such as Anti-Virtual Machine Detection and Process Injection to evade security tools.
- These techniques make malware analysis more challenging and highlight the importance of advanced detection methods.

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Conclusion

- Detection-evasion techniques are commonly used by modern malware to avoid antivirus software and sandbox environments.
- Understanding these techniques enables cybersecurity professionals to recognize sophisticated threats and implement stronger security measures. Combining static analysis, dynamic analysis, and behavioral monitoring significantly improves malware detection and analysis.